P. SRI SAI CHARAN

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EDUCATION

Vardhaman College of Engineering, Hyderabad Aug 2024 – May 2028
B.Tech, Computer Science (Artificial Intelligence & Machine Learning) • CGPA: 8.45 / 10

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Backend & Systems: FastAPI, async SQLAlchemy, Alembic migrations, Celery, REST API design, Pydantic, static analysis (tree-sitter), graph algorithms, threat modeling, rate limiting
Data & Infrastructure: PostgreSQL, Qdrant (vector search), Redis, SQLite, schema design, Docker & Docker Compose, GitHub Actions CI, Caddy/nginx
ML & Retrieval: RAG pipelines, embeddings, sentence-transformers, scikit-learn, pandas, NumPy, TF-IDF, Random Forest, offline benchmarking (precision@k, MRR)
Frontend: React 18, Next.js, TypeScript, Vite, Tailwind CSS, PWA
Testing & Tools: pytest (unit + integration), Git, load testing, Jupyter, Postman

ENGINEERING PROJECTS

CodeLens — Code Knowledge Graph & Change-Impact Analysis 56 commits • 17.5K LOC • 290 tests
Python • FastAPI • tree-sitter • Next.js • TypeScript • PostgreSQL • Docker • Caddy

- Architected a static-analysis service that parses arbitrary repositories with tree-sitter, resolves cross-file call and import edges, and exposes a three-level navigable graph (architecture → module → symbol) across 59 REST endpoints.
- Built a blast-radius query computing the reverse transitive closure from any node, ranked on four normalized signals — graph distance, git co-change strength, churn, and structural fan-in — returning the dependency path that substantiates every result.
- Established a reproducible accuracy benchmark using multi-file git commits as ground truth, grading 1,203 examples across 8 open-source repositories; published precision@10 of 0.232 against a 0.221 popularity baseline, alongside two earlier entries that measured below baseline.
- Modeled edge confidence explicitly (resolved / heuristic / dynamic_unknown) so unresolved dynamic dispatch renders as uncertainty in the UI rather than as fact.
- Audited the untrusted-input surface — anonymous repository URLs cloned and parsed server-side — closing SSRF via a URL allowlist and enforcing in-flight disk, memory, CPU and concurrency ceilings; covered by 290 tests across 25 modules.

SUSTAIN — Multi-Tenant Clinical Decision Support System 39 endpoints • 23 tests
Python • FastAPI • PostgreSQL • Qdrant • Celery • Redis • React • Docker Compose

- Designed a multi-tenant CDSS backend on FastAPI with async SQLAlchemy, Alembic-managed migrations, and Celery/Redis scheduled jobs, spanning consultation, prescription, follow-up, surveillance and audit domains.
- Engineered a RAG retrieval pipeline over PostgreSQL and Qdrant that degrades across three tiers — file-backed vector store without Qdrant, hashing embedder without sentence-transformers, raw retrieved evidence without an LLM key — so the full clinical flow runs on Python and Postgres alone.
- Implemented a tamper-evident audit log chaining every clinical decision by SHA-256 over the previous hash plus event payload, making retroactive modification of any earlier record detectable.
- Built a rule-based recovery-surveillance engine classifying patient trajectories into a five-category anomaly taxonomy with triage severities, adaptively rescheduling the next check-in between 1 and 5 days from the observed trend.
- Enforced tenant isolation under integration test and wired CI on every push — Postgres service container, pytest, frontend type-check and build.

CareerLens — Resume-to-Roadmap Analysis Pipeline 27 commits • 44 endpoints • 93 tests
Python • FastAPI • PostgreSQL • React 18 • TypeScript • Docker Compose

- Built a seven-stage resume-to-roadmap pipeline with progressive per-stage rendering, backed by FastAPI and PostgreSQL and deployed via Docker Compose.
- Designed an explainable hybrid scorer combining TF-IDF text similarity, skill-graph proximity, and market-demand signal, exposing each component's contribution instead of a single opaque score.
- Diagnosed and fixed a cross-session data leak in which a logged-out user's in-flight responses could render into the next session, and made re-running any stage invalidate all downstream derived results.

ACHIEVEMENTS

- **WaveX (team hackathon project):** contributed to a multi-agent decision-intelligence platform in a shared repository — onboarding flow, advisor context injection, and a metrics/BI layer rendering structured advisor output as charts.
- **LeetCode:** 105+ problems solved (33 Medium, 6 Hard).